

# Doyeon (Doy) Kim

QUANTITATIVE RESEARCHER · PROGRAM EVALUATION · EDUCATIONAL TECHNOLOGY

Madison, WI

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## Summary

Learning scientist and quantitative researcher with a decade of experience designing and evaluating education programs and technologies in authentic K–12 and postsecondary settings. Currently a Doctoral Research Intern in WestEd’s Center for Teaching and Learning, building and maintaining end-to-end data pipelines and evaluating educational technology interventions through randomized and quasi-experimental designs. Background spans rigorous quantitative methods (experimental design, structural equation modeling, causal inference), mixed-methods data collection, and technical reporting for diverse audiences. Former certified secondary mathematics teacher and developer of mathematics professional development for over 7,000 in-service teachers.

## Relevant Experience

### WestEd, Center for Teaching and Learning

Remote

DOCTORAL RESEARCH INTERN

Jun 2023–Present

- Evaluate the efficacy of educational technologies for high-need and rural students through randomized and quasi-experimental designs, including a school-level cluster RCT pre-registered to meet What Works Clearinghouse (WWC) standards
- Build and maintain end-to-end data pipelines integrating multi-source administrative, survey, platform-usage, and assessment data; perform systematic cleaning, harmonization, and quality assurance across longitudinal teacher and student datasets
- Develop instruments and lead data collection through interviews, online surveys, and site visits; conduct analyses and contribute to client reports and dissemination products
- Gather assessment data from state education agencies and visualize data-collection progress to monitor study quality and timeline

### University of Wisconsin-Madison

Madison, WI

GRADUATE RESEARCHER (LEARNING SCIENCES)

Aug 2019–Present

- Design and conduct randomized experiments on mathematics learning in technology-enhanced environments; apply structural equation modeling, item response theory, and integrative data analysis across multiple datasets
- Founded and lead the iGRASP research team, supervising 1 graduate and 6 undergraduate research assistants in experiment administration, scientific data collection, and project coordination
- Pioneered a remote data-collection protocol during COVID-19; present findings at national conferences and in peer-reviewed proceedings

### University of Wisconsin-Madison

Madison, WI

TEACHING ASSISTANT

Aug 2021–Aug 2025

- Teaching assistant for EDPSYCH 301 (How People Learn); delivered instruction, feedback, and individualized support on learning theory to undergraduate students

### Center for In-Service Teacher Education

Seoul, South Korea

GENERAL MANAGER

Jun 2016–Oct 2017

- Managed 50+ professional development programs (1,500+ hours) reaching over 7,000 in-service teachers; collected program feedback through surveys
- Raised the institutional evaluation grade from B to A across the 2016–2018 review cycle

### EBS Math, Educational Broadcasting System

Seoul, South Korea

EDUCATIONAL CONTENT RESEARCHER

Apr 2013–Jun 2014

- Developed 44 curriculum-aligned instructional plans and produced 77 video and 55 animated episodes translating mathematics content for broad public audiences

## Skills

|                                                                                                                                                                                                            |                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Prompt engineering for research & technical tasks · AI-assisted literature review, visualization & statistical programming · Output verification, bias awareness & code review                             | Generative AI                          |
| Field research coordination & participant recruitment · Research team supervision & mentorship · Multi-project prioritization under deadlines                                                              | Project Management                     |
| Interviews, focus groups & surveys · Observation & site visits · Discourse & video analysis · Epistemic Network Analysis (ENA); qualitative coding (MAXQDA)                                                | Qualitative & Mixed Methods            |
| Experimental & quasi-experimental design · Causal inference · Structural equation modeling · Hierarchical linear modeling · Item response theory & measurement · Meta-analysis & integrative data analysis | Quantitative Methods                   |
| Technical & evaluation report writing · Data visualization (charts, tables, graphics) · Practitioner-oriented materials & professional development · Presentation & meeting facilitation                   | Reporting & Communication              |
| R (statistical programming) · End-to-end data pipeline construction · Data cleaning, harmonization & quality assurance · Multi-source data integration (administrative, survey, platform, assessment)      | Statistical Software & Data Management |

## Education

### University of Wisconsin-Madison

PH.D. IN EDUCATIONAL PSYCHOLOGY – LEARNING SCIENCES

- Minor: Quantitative Educational Research
- Advisor: Mitchell J. Nathan

Madison, WI

Jan 2023–Present

### University of Wisconsin-Madison

M.SC. IN EDUCATIONAL PSYCHOLOGY – LEARNING SCIENCES

- Advisor: Mitchell J. Nathan

Madison, WI

Aug 2019–Dec 2022

### Seoul National University

M.SC. IN MATHEMATICS EDUCATION

- Coursework Emphasis: Cognitive Science
- Advisor: Oh Nam Kwon

Seoul, South Korea

Mar 2016–Feb 2019

### Seoul National University

B.SC. IN MATHEMATICS EDUCATION

- Advisor: Oh Nam Kwon

Seoul, South Korea

Mar 2009–Feb 2016

## Selected Publications

1. Kim, D., Schenck, K. E., Walkington, C., & Nathan, M. J. (2026). Embodied cognitive pathways from dynamic geometry software access to deductive geometric reasoning. *Proceedings of the 20th International Conference of the Learning Sciences (ICLS 2026)*.
2. Kim, D., Kim, C., & Nathan, M. J. (2026). What makes actions matter in virtual reality? Examining the effects of conceptual congruence and sensorimotor engagement on geometric reasoning. *Proceedings of the 12th Immersive Learning Research Network Conference (iLRN 2026)*.
3. Kim, D., Beliakoff, A., & Strother, S. (2025). Impact evaluation of a virtual reality program on students' geometry achievement. *SREE 2025 Conference: Education in Context – Research, Systems, and the Future of Evidence-Based Change*.
4. Kim, D., Schenck, K. E., Xia, F., Swart, M. I., Walkington, C., & Nathan, M. J. (2025). The effect of dynamic geometry simulation access on embodied geometric reasoning: An experiment. *2025 AERA Annual Meeting*.
5. Park, W., Kim, D., & Kang, D. Y. (2021). Research trends in science and mathematics education in South Korea 2014–2018: A cross-disciplinary analysis of publications in selected local journals. *Asia-Pacific Science Education*, 7(2), 280–308.